Predictive Quality Classification in Additive Manufacturing with a hybrid approach

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Research ‘problem’ or justification

The quality challenge in additive manufacturing

- Rise of additive manufacturing

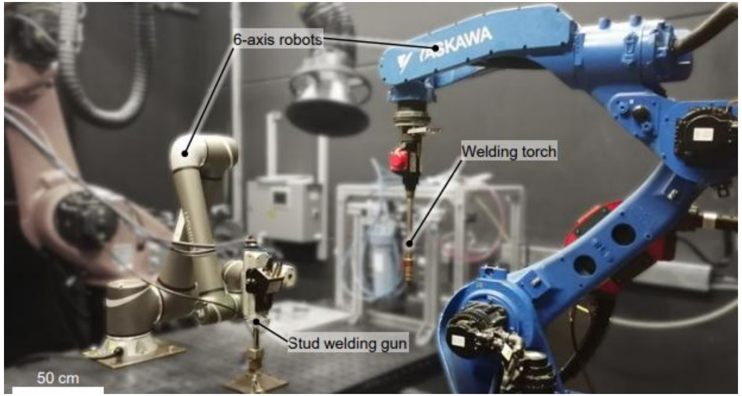
  Handle growing complexity and customization

- Quality uncertainty

  Process variability and limited post-process inspection

- Real-time quality control need

  To ensure the robustness of the S-WAAM process



Research ‘problem’ or justification

Predictive modeling via sensors

- In-situ data collection

  Using sensors during the process.

- Key variables

  Current, Voltage, Acoustic Emissions.

- Goal : classification model

P1000-S3 sensor

Welding microphone
(Xiris Automation GmbH)

Gap in the litterature

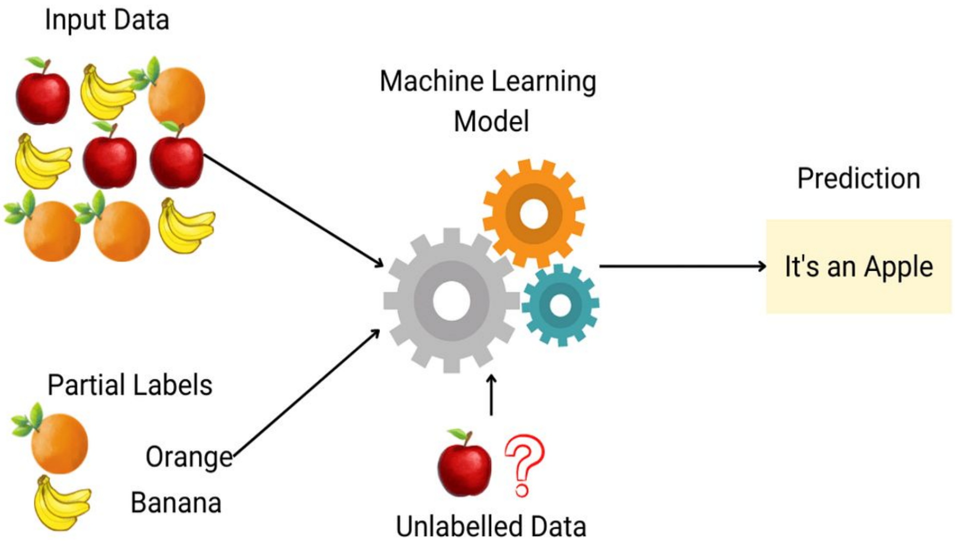
Current Research Focus:

- Most previous studies on *Wire Arc Additive Manufacturing (WAAM)* anomaly detection rely on **supervised learning** models, which require large, well-labeled datasets of both normal and defective data.
- However, **defective data are rare and difficult to collect** in industrial WAAM settings because certification processes usually record only *defect-free* depositions.
- **Unsupervised learning** approaches have been explored as a solution, but they often **underperform** compared to supervised methods and lack robustness for real-time applications.

Gap in the litterature

Identified Gaps in the Literature:

- Lack of **semi-supervised** methods that can operate effectively using *only good (non-defective)* data.
- Insufficient exploration of **time-frequency domain analysis** for feature extraction—most research still relies on time-domain metrics.
- Absence of **real-time** anomaly detection frameworks aligned with industrial standards.

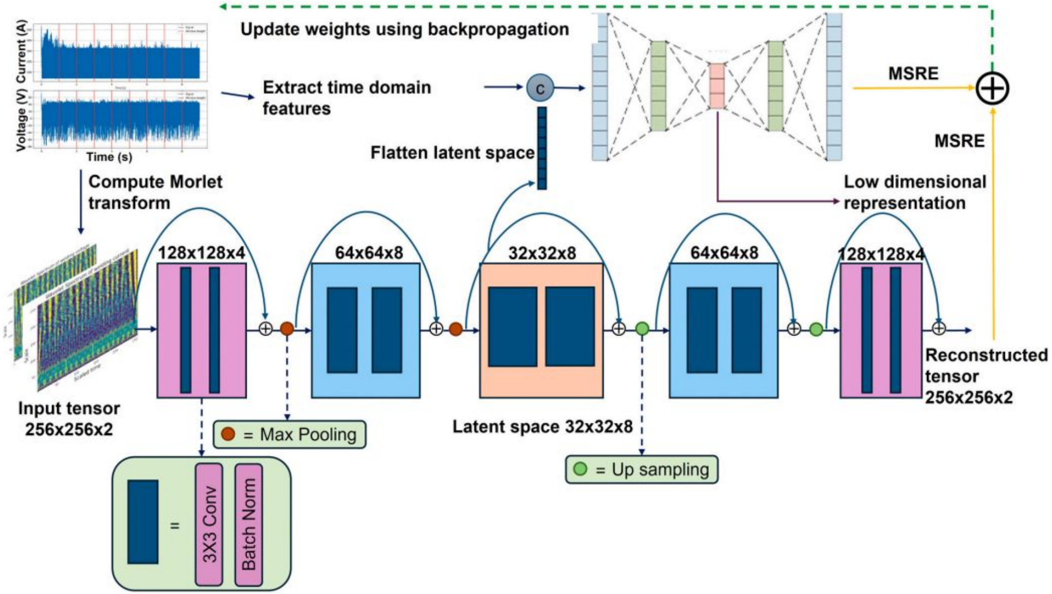


*Semi-supervised method explanation, source: https://miro.medium.com/v2/resize:fit:1200/1*snZhMEQFhoJwbM5c0CPOAw.png*

Gap in the litterature

Contribution of the Paper:

- Introduces a **semi-supervised framework** using a **Double Residual Convolutional Autoencoder (CAE)** trained exclusively on *defect-free* data.
- Utilizes **Morlet Continuous Wavelet Transform (CWT)** to extract **local time-frequency features**, capturing transient instabilities that traditional time-domain models miss.



The proposed Double Residual Convolutional Autoencoder (Fig. 13, Mattera et al., 2024) combines time- and frequency-domain features for anomaly detection

Research Aim

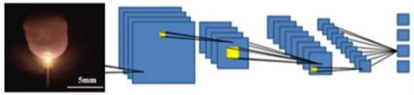
Classification model

- hybrid data: audio signals + welding current and voltage signals

State of the art of anomaly detection in WAAM

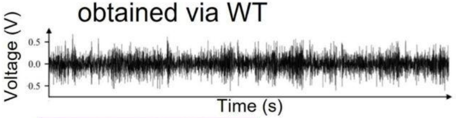
Welding pool images

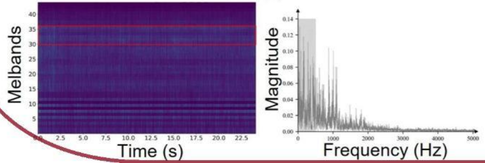
- Features extraction using image processing
- Direct analysis via Convolutional Neural Networks



Audio signals

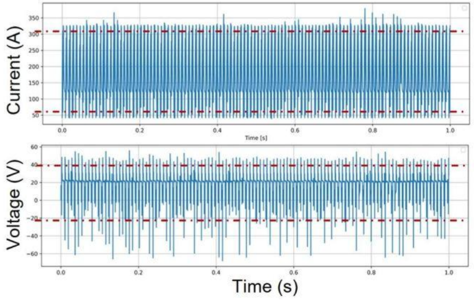
- Features extraction in time and frequency domain
- Analysis of the spectrogram obtained via FFT
- Analysis of the scalogram obtained via WT





Welding current and voltage signals

- Features extraction in time domain



The Dataset

Total samples

260

iO passed

niO failed

Total usable samples

~ 200

after data cleansing

Data characteristics:

216 welding records

voltage and current data

201 audio files

acoustics measurements

- pre-labeled binary classification (iO/niO)
- combination of welding data and audio data

Process Parameters:

current: 390 A

time: 60ms

penetration depth: 2 mm

stroke: 2 mm

Objectives

1. Evaluating model performance on small dataset
2. Compare multiple ML models for determining optimal for DASW quality prediction
3. Optimize feature selection for iO/niO classification

Questions

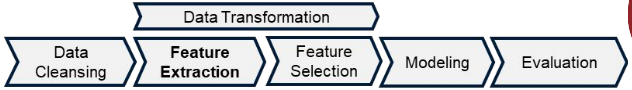
Data exploration - What are the characteristics of DASW data? Are there any data quality issues? Are there distinctive patterns between iO and niO parts?

Feature engineering - Which minimal combination of audio and current/voltage features achieves the best distinguishes good and defective parts?

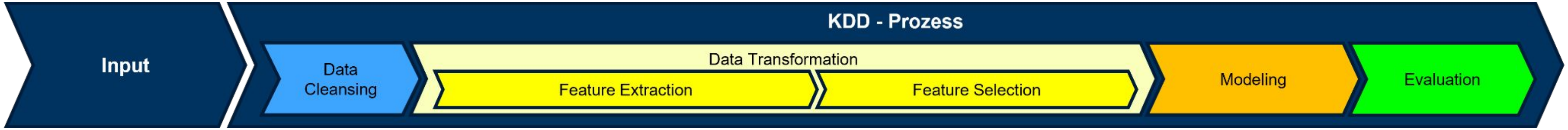
Modeling - Which model it the best for our data?

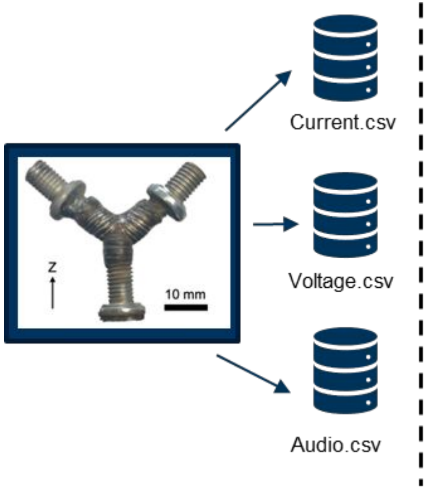
Model evaluation - How well the model performs? What are the sources of error?

Research method & methodology



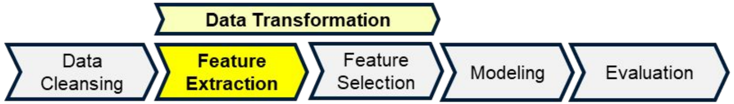
Realisation of the project along the **Knowledge Discovery in Database (KDD)** Process

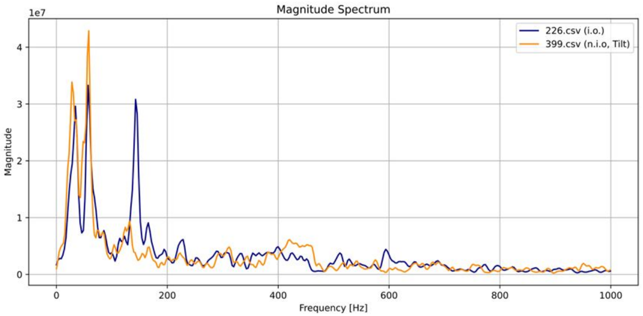


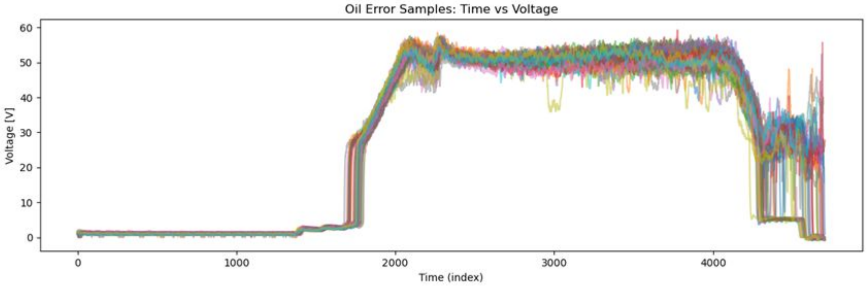


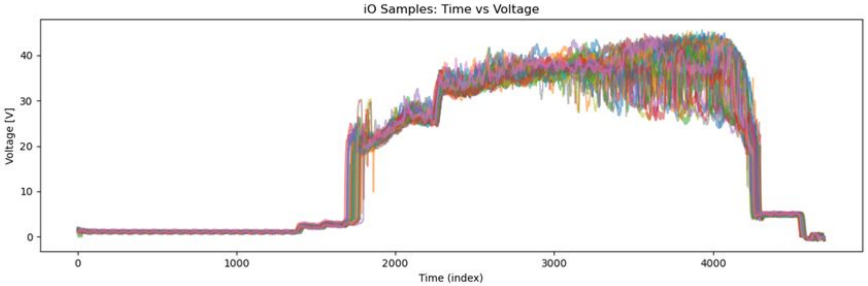
- outliers
- missing values
- inconsistency

Feature Extraction

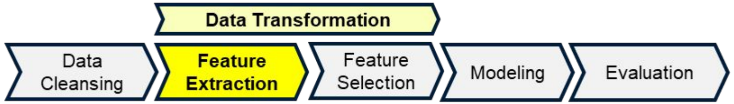


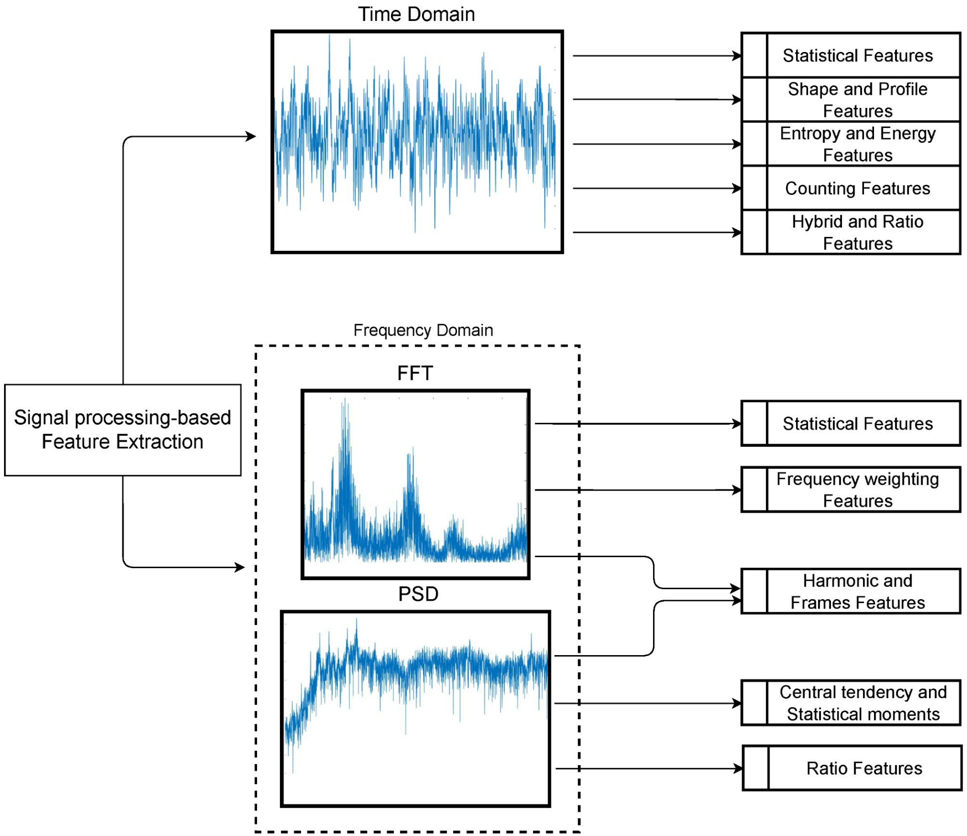






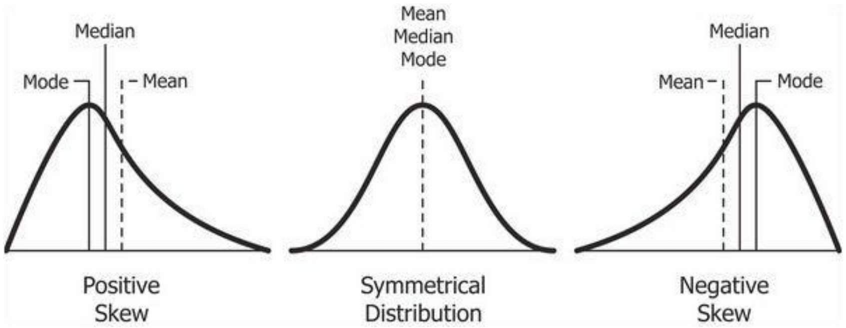
Feature Extraction





Audio, Current/Voltage Feature Extraction

- Preprocessing: filtering, segmentation into relevant windows, normalization
- Extraction of Statistical, temporal, spectral Features
- E.g. Mean, Standard Deviation, Skewness, Kurtosis, Peak-to-Peak value



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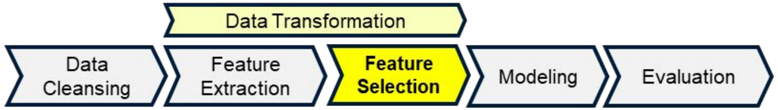
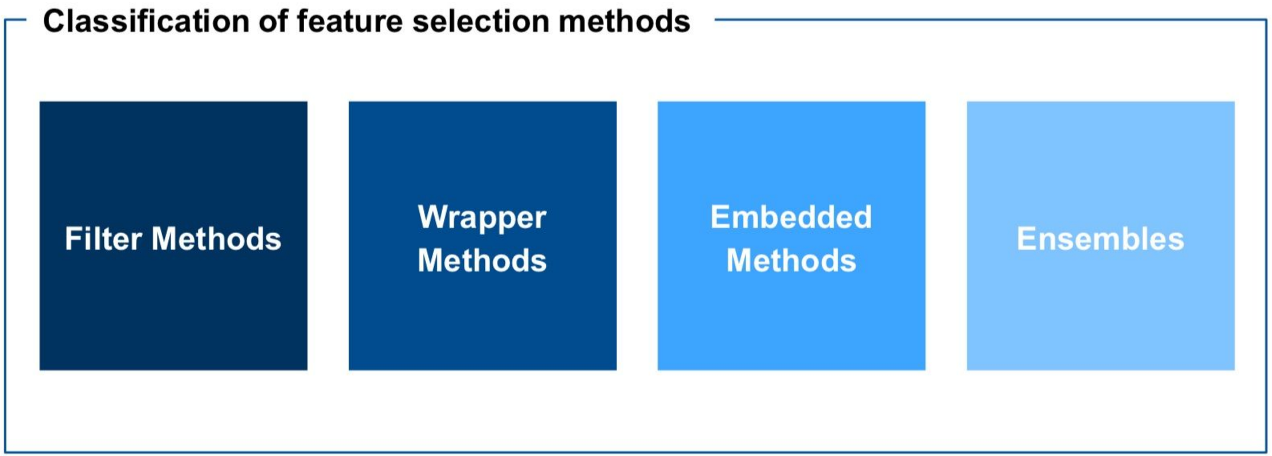
Feature Selection



- decision making process which features are used as model input
- Combining Features of both, Audio and Current/Voltage measurements
- Visualizing Features in a combined feature space e.g. PCA or 2D plots

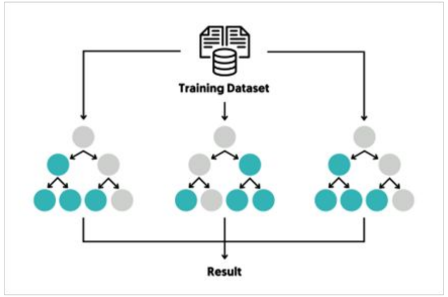


Modeling

Data Transformation

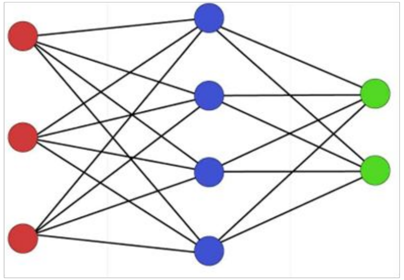
Data Cleansing | Feature Extraction | Feature Selection | **Modeling** | Evaluation

Random Forest Classifier



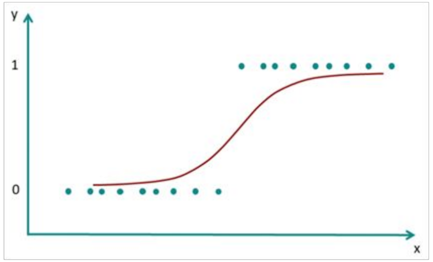
https://dida.do/de/was-ist-ein-random-forest

Neuronales Netzwerk



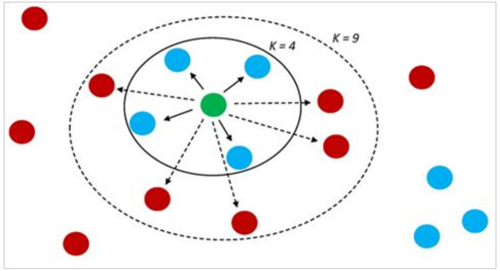
https://datasolut.com/neuronale-netzwerke-einfuehrung/

Logistic Regression



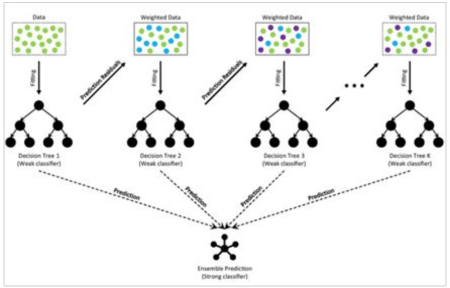
https://datatab.de/tutorial/logistische-regression

K-Nearest Neighbor



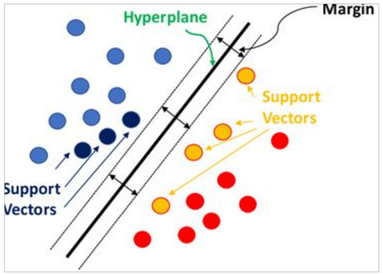
https://www.researchgate.net/figure/k-nearest-neighbors-A-diagram-showing-an-example-of-the-k-nearest-neighbor-machine_fig1_356781515

Gradient Boosting



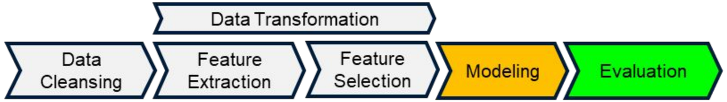
https://www.researchgate.net/figure/The-architecture-of-Gradient-Boosting-Decision-Tree_fig2_356698772

Support Vector Machine (RBF Kernel)



https://medium.com/@rizwan44007/mastering-support-vector-machines-svms-f45c0d9eb33c

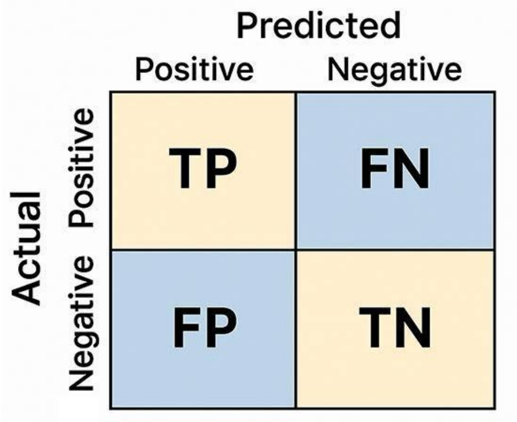
Model Evaluation



Evaluation and Comparison of Models through common metrics:

- Accuracy Scores
- F1-Score
- Precision
- Recall

Confusion Matrix



https://statisticsbyjim.com/glossary/confusion-matrix/

Summary and Proposed Outcomes

The Challenge ?	Our Hybrid Soluti	Impact & Outcome
●Quality in Wire Arc Additive Manufacturing (WAAM) is uncertain. ●Defective data are rare, making supervised training difficult. ●Real-time quality monitoring is needed for robust processes.	●Use sensor data: audio + current/voltage signals. ●Extract time-domain (audio) and statistical (current/voltage) features. ●Combine features for supervised classification.	●Reliable quality prediction with standard metrics (Accuracy, F1, Precision, Recall, Confusion Matrix). ●Identify the minimal effective feature set. ●Demonstrate that hybrid data improves prediction vs single-source signals.